

Aryan Naveen Singh Bhutyal

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EDUCATION

SRM Institute of Science and Technology

Chennai, India

Bachelor of Technology in Computer Science and Engineering — CGPA: 9.72/10

2023 – 2027

Relevant Coursework: Data Structures and Algorithms, Operating Systems, Database Management Systems, Computer Networks, Machine Learning

EXPERIENCE

PayU

Feb 2026 – Apr 2026

Research Intern

Remote

- Engineered an Agentic Fraud Risk Management platform using the Model Context Protocol (MCP) to orchestrate 13+ investigation tools.
- Built a two-stage fraud detection pipeline: real-time triage followed by AI-assisted investigation for high-risk transactions.
- Integrated XGBoost and SHAP for explainable fraud predictions and analyst-friendly feature attribution.
- Engineered real-time velocity and IP intelligence features to improve detection while reducing false positives.

Samsung R&D Institute Bangalore (Samsung PRISM)

Mar 2025 – Aug 2025

Research Intern

Remote

- Led an AI-assisted SQL optimization system evaluated across 30+ analytical queries and multiple database schemas.
- Built a schema intelligence pipeline with a 3-member team using SQLAlchemy, supporting MySQL, PostgreSQL, and SQLite.
- Designed an execution-aware profiling workflow to empirically compare SQL rewrites.
- Developed a Flask backend and React frontend for side-by-side SQL optimization analysis.

Indian Institute of Technology (IIT) Jammu

Dec 2024

Machine Learning Intern

Jammu, India

- Applied Logistic Regression, XGBoost, and K-Means clustering on structured datasets for predictive modeling and pattern discovery.
- Improved model performance through feature engineering, cross-validation, and hyperparameter tuning.

PROJECTS

• ClinXplain – Ontology-Aware Clinical Decision Support System

Sept 2025 – Mar 2026

- Developed an explainable clinical risk prediction platform for cardiovascular disease and Type 2 diabetes using ML and medical ontologies.
- Evaluated models on 67k+ patient records with India-specific post-hoc risk calibration.
- Combined SHAP explanations with a Neo4j knowledge graph linked to SNOMED-aligned clinical findings.

• Enhanced Deep Learning Ensemble for Malaria Classification

Aug 2025 – Nov 2025

- Built a three-model CNN ensemble with patient-wise 80/10/10 validation, achieving 97.18% accuracy on the NIH malaria dataset.
- Integrated Grad-CAM visualization to improve model interpretability and parasite localization.

• Real-Time Face Mask Detection System

Apr 2025 – May 2025

- Implemented a CNN-based real-time face mask detection pipeline using TensorFlow, Keras, and OpenCV, achieving 92% accuracy at over 15 FPS.

TECHNICAL SKILLS

• **Programming Languages:** C++, C, Python, SQL

• **Frameworks & Libraries:** Flask, Scikit-learn, Pandas, NumPy, Matplotlib, OpenCV, Keras, SHAP

• **Databases:** MySQL, PostgreSQL, SQLite, Neo4j

• **Core Computer Science:** Data Structures and Algorithms, Object-Oriented Programming, Operating Systems, Database Management Systems

• **Tools:** Git, Jupyter Notebook, Google Colab, Neo4j